Taehwan Jin, *Ph.D.*

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Google Scholar: https://scholar.google.com/citations?user=Zy2s_K4AAAAJ
GitHub: https://taehwan-jin.github.io
Nationality: South Korea
Languages: Korean (native), English (high intermediate)
Korean Residence (Current): Seongnam-si, Gyeonggi-do, South Korea
Swedish Residence: Solna, Stockholm, Sweden
Type of permit in Sweden: Residence and work permit (UPPEHÅLLS- OCH ARBETSTILLSTÅND)

PROFESSIONAL SUMMARY

Seasoned analog/RF IC design engineer with over 10 years of experience developing high-performance LNAs, mixers, and transimpedance amplifiers for mobile and optical communication applications. Led cross-functional teams to deliver 200 Gbps/4-ch PAM4 optical transceivers, and contributed to RF front-end ICs powering Samsung Galaxy flagship smartphones. Extensive hands-on experience across Samsung Foundry nodes (28nm planar FET, 14nm/8nm/5nm/4nm FinFET, 2nm GAAFET) and SiGe BiCMOS, with proven proficiency in Cadence Virtuoso from schematic to tape-out, including post-layout verification.

PROFESSIONAL EXPERIENCE

QUALITAS SEMICONDUCTOR, CO., LTD., SEONGNAM, SOUTH KOREA

Principal Engineer, IP Development Team *(Jan. 2025 – Present)*

- Led MIPI Taskforce, orchestrating cross-functional team initiatives
- Managed MIPI D/C-PHY IP Design service tasks, ensuring on-time milestones and quality standards
- Coordinated with layout, analog design, digital design, and application engineering groups to ensure seamless IP integration
- Established Analog-Mixed Signal simulation environment and performed test vector verification
- Designed analog circuits (T-coils, DC reference voltage cleaner)

Principal Engineer, Optical Interconnect Team *(Apr. 2021 – Jan. 2025)*

- Led Optical Front-End IC Design group
- Coordinated cross-functional groups across layout, optical packaging, device modeling, and digital design
- Developed optical transceiver ICs (VCSEL driver IC, TIA IC)
- Conducted optical link budget analysis to ensure system-level performance compliance
- Verified IC top and designed analog circuits (high-speed TIA, POR, BGR)

SAMSUNG ELECTRONICS, CO., LTD. (System LSI Division), HWASEONG, SOUTH KOREA

Senior Engineer, RF Development Team *(Mar. 2015 – Mar. 2021)*

- Designed RF front-end receiver circuits for Samsung Galaxy flagship smartphones
- Served as designer for cellular receiver IP top and front-end sub-blocks

Project Timeline

2025

- Qualitas Semiconductor / IP Development Team / Leader: Yong Hyun Park
- **MIPI Taskforce Leader** *(Current Role)*
    - Managed MIPI D/C-PHY IP design service tasks, orchestrated schedule management, and task allocation
    - Performed Analog-Mixed Signal simulation for test vector verification
    - Executed MPW for Qualitas's proprietary licensing IP development
    - *Process:* 14nm, 8nm, 5nm, 4nm FinFET, 2nm GAAFET
    - *Customers:* Ambarella, Visionnext, KeelinMicro, HHXmicro, Telechips, Axis, etc.
- **Design Engineer, IP Development Team** *(Early 2025)*
    - Designed T-coils for input/output matching for High-Speed Analog Front-End (AFE)
    - Designed DC reference voltage cleaning circuits for Current-Controlled Oscillators (CCO)
    - *Application:* Multi-Standard SerDes
    - *Process:* Samsung Foundry 4nm FinFET CMOS (SF4)

2021–2024

- Qualitas Semiconductor / Optical Interconnect Team / Leader: Jae Young Kim, *Ph.D.*
- Led Optical Front-End IC Design group; verified IC top and designed high-speed optical front-end receiver (TIA) IC
    - QSOD250x (Engineering Sample)
        - 100 Gbps/4-Channel NRZ SiGe BiCMOS Optical Front-End Transceiver ICs
        - Application: Active Optical Cable for HDMI/DisplayPort combo solutions
        - Process: Tower Semiconductor, SiGe BiCMOS 180nm (TS_SBC18S5)
        - Package: COB & Optical Sub-Assembly
    - IOET050x (Government-funded R&D Project, MPW)
        - 200 Gbps/4-Channel PAM4 CMOS Optical Front-End Transceiver ICs
        - Application: Active Optical Cable for data center
        - Process: 14-nm FinFET CMOS (SF_LN14LPU)
        - Package: COB & Optical Sub-Assembly

2020

- Samsung Electronics Co. (S.LSI) / Mixed Signal Circuit Design Team / Leader: Jong Woo Lee, *Ph.D.*
- Designed mid- and high-band LNA
    - S5520 (RFIC for Galaxy S22)
        - Process: 14nm FinFET CMOS RF (SF_LNR14LPP)
        - Package: FC-FBGA

2019

- Samsung Electronics Co. (S.LSI) / Mixed Signal Circuit Design Team / Leader: Jong Woo Lee, *Ph.D.*
- Designed loopback circuit for IIP2 calibration between TX and RX paths
    - S5500 (RFIC for Galaxy S10)
        - Process: 14nm FinFET CMOS RF (SF_LNR14LPP)
        - Package: FC-FBGA

2018

- Samsung Electronics Co. (S.LSI) / Infra Design Team / Leader: Thomas Byunghak Cho, *Ph.D.*
- Served as receiver IP top; Designed low-band LNA
    - S938 (RFIC for Galaxy Tab S6 Lite)
        - Process: 28nm CMOS RF (SF_LNR28LPP)
        - Package: FC-FBGA

2017

- Samsung Electronics Co. (S.LSI) / RF Development Team / Leader: Thomas Byunghak Cho, *Ph.D.*
- Designed ultra-high-band LNA
    - S965 (RFIC for Galaxy S9)
        - Process: 14nm FinFET CMOS RF (SF_LNR14LPP)
        - Package: WLP

2016

- Samsung Electronics Co. (S.LSI) / RF Development Team / Leader: Thomas Byunghak Cho, *Ph.D.*
- Designed down-conversion mixer and baseband TIA for 3G/LTE
    - S955 (RFIC for Galaxy S8 plus)
        - Process: 14nm FinFET CMOS RF (SF_LNR14LPP)
        - Package: WLP

2015

- Samsung Electronics Co. (S.LSI) / RF Development Team / Leader: Thomas Byunghak

Cho, *Ph.D.*
- Designed low-band balun for receiver for 3G/LTE
  - S935 (RFIC for Galaxy S7)
    - Process: 28nm CMOS RF (SF_LNR28LPP)
    - Package: WL-CSP
- Designed active balun LNA for GNSS, Glonas, Baidu, Galileo
  - S955 (RFIC for Galaxy S8 plus)
    - Process: 14nm FinFET CMOS RF (SF_LNR14LPP)
    - Package: WLP

EDUCATION

YONSEI UNIVERSITY, SEOUL, SOUTH KOREA

Ph.D. in Electrical and Electronic Engineering *(February 2015)*

- Combined M.S./Ph.D. Program
- Dissertation: *Time-of-Arrival Measurement Using Adaptive CMOS Impulse Radio UWB Range Finder with Scalable Resolution*
- Advisor: Prof. Tae-wook Kim, *Ph.D.*

Research Assistant, Circuits and Systems Design Laboratory *(Mar. 2008 – Feb. 2015)*

- High-Linearity Low-Noise Amplifier Design and Analysis
  - Designed Multiple Gated Transistor LNA in 180-nm CMOS; analyzed linearity and noise performance, achieving +9.4 dBm IIP3
  - Designed and implemented bulk-biased MGTR LNA achieving +12.5 dBm IIP3 in 180-nm CMOS; described its operating principle
  - Designed and implemented frequency-selective shunt feedback LNA in 180-nm CMOS; analyzed linearity and noise performance, achieving +8.2 dBm IIP3
- CMOS IR-UWB Range-Finding Transceiver with TDC-Based Time-of-Flight (ToF) Measurement
  - Designed and implemented adaptive IR-UWB transceiver system with scalable resolution in 110-nm CMOS
    - Key blocks: LNA, Time-to-Digital Converter, Envelope Detector, Comparator, Digital Impulse Generator
    - Demonstrated multipath-resistant range detection up to 8 m with 8.8 mm resolution
    - Performed leading-edge detection system-link budgeting
    - Established range measurement test infrastructure

Visiting Scientist, IHP Microelectronics, Frankfurt (Oder), Germany *(Nov. 2011 – Mar. 2012)*

- Investigated impulse-radio ultra-wideband systems for indoor localization applications

- Designed low-noise amplifier receiver front-end using 0.25-μm SiGe BiCMOS technology
- Characterized highly optimized on-chip transformers for wideband input matching in LNA design

KWANGWOON UNIVERSITY, SEOUL, SOUTH KOREA

B.S. in Electronic Engineering *(February 2008)*

KEY QUALIFICATIONS

EDA Tools

- **Circuit Design & Simulation:** Cadence Spectre/Virtuoso (ADE L, Maestro), ADS/ Momentum, FINESIM, Hspice, StarRC, PEX, Primesim, VCS
- **Design & Process Verification:** DRC, LVS, TOTEM, PERC, BCRSIM
- **EM Field Solver & PCB:** EMX/EMXD, PADS

Measurement Equipment Experience

Vector Network Analyzer, Noise Figure Analyzer, Spectrum Analyzer, Vector Signal Analyzer, Logic Analyzer, Oscilloscope, Signal Generator, Probe Station

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PUBLICATIONS

1. Youngmin Kim, Junghwan Han, Jae-Seung Lee, **Taehwan Jin**, Pilsung Jang, Heeseon Shin, Jongwoo Lee, and Thomas Byunghak Cho, "Power-Efficient CMOS Cellular RF Receivers for Carrier Aggregation According to RF Front-End Configuration," *IEEE Trans. Microwave Theory and Technology*, vol. 69, no. 1, pp. 452–468, Jan. 2021.

2. **Taehwan Jin**, Hong Gul Han, and Tae Wook Kim, "Time-of-Arrival Measurement Using Adaptive CMOS IR-UWB Range Finder with Scalable Resolution," *IEEE Trans. Circuits and Systems I, Reg. Papers*, vol. 63, no. 10, pp. 1605–1615, Oct. 2016.

3. **Taehwan Jin**, Hong Gul Han, and Tae Wook Kim, "A 0.7-dB NF, +8.2-dBm IIP3 CMOS Low Noise Amplifier Using Frequency Selective Feedback," *International Journal of Circuit Theory and Applications*, vol. 44, no. 1, pp. 21–37, Jan. 2015.

4. **Taehwan Jin** and Tae Wook Kim, "A 6.75 mW +12.45 dBm IIP3 1.76 dB NF 0.9 GHz CMOS LNA Employing Multiple Gated Transistors with Bulk-Bias Control," *IEEE Microwave and Wireless Components Letters*, vol. 21, no. 11, pp. 616–618, Nov. 2011.

5. **Taehwan Jin** and Tae Wook Kim, "A 5.5 mW +9.4 dBm IIP3 1.8 dB NF CMOS LNA Employing Multiple Gated Transistor with Capacitance Desensitization," *IEEE Trans. Microwave Theory and Technology*, vol. 58, no. 10, pp. 2529–2537, Oct. 2010.

CONFERENCE PRESENTATIONS

1. Junseo Kim, Kihun Kim, **Taehwan Jin**, Pyungsu Han, and Wooyoung Choi, "4 × 50 Gb/s 0.85 pJ/bit PAM-4 CMOS VCSEL Driver for Linear Pluggable Optics," *Conference on Lasers and Electro-Optics/Pacific Rim (CLEO-PR)*, Optica Publishing Group, p. Mo1K_3, Aug. 5, 2024.

2. Youngmin Kim, Pilsung Jang, **Taehwan Jin**, Jae-Seung Lee, Heeseon Shin, Suseob Ahn, Jungyeol Bae, Junghwan Han, Seungchan Heo, and Thomas Byunghak Cho, "A Current-Efficient Wideband Cellular RF Receiver for Multi-Band Inter- and Intra-Band Carrier Aggregation Using 14nm FinFET CMOS," *IEEE Radio Frequency Integrated Circuits Symposium (RFIC)*, Jun. 4–6, 2017.

3. **Taehwan Jin**, Hong Gul Han, and Tae Wook Kim, "Time-of-Arrival Measurement Using Adaptive CMOS IR-UWB Range Finder with Scalable Resolution," *IEEE International Symposium on Circuits and Systems (ISCAS)*, Baltimore, MD, USA, May 28–31, 2017.

4. **Taehwan Jin** and Tae Wook Kim, "A Highly Linear LNA Employing Transconductance Nonlinearity Cancellation with the Desensitization Technique of Harmonic Feedback Effect," *International Conference on Electronics, Information, and Communication (ICEIC)*, Tashkent, Uzbekistan, June 23-26, 2008.

PATENTS

5 registered patents out of 18 filed applications

1. **KR 10-2875615(2025.10.20)**
   Optical transmitter and optical receiver for optical communication
   Jae-Young Kim, **Tae-Hwan Jin**, Jun-seo Kim, Hyoung-Jun Na
   Qualitas Semiconductor

2. **KR 10-2540949(2023.06.01) / US 11671292 B2(2023.06.06)** *A signal processing circuit improving linearity of pulse amplitude modulated signal and communication device including the circuit*
   Tae-Hwan Jin, Jae-Young Kim, Du-Ho Kim
   Qualitas Semiconductor

3. **US 11,005,426 B2(2021.05.11) / EP 3550721 B1(2023.03.15)** *Radio frequency (RF) integrated circuit performing signal amplification operation to support carrier aggregation and receiver including the same*
   Young-Min Kim, Jong-Soo Lee, Jong-Woo Lee, Joong-Ho Lee, Ji-Young Lee, Pil-Sung Jang, Thomas Byunghak Cho, **Tae-Hwan Jin**
   Samsung Electronics Co., Ltd.

4. **US 10,250,195 B2(2019.04.02) / KR 10-2504292(2023.02.28) / CN 108075794(2021.02.09)** *Receiver receiving wideband radio frequency signal, wireless communication device including the same, and method of operating the wireless communication device*
   Young-Min Kim, Jae-Seung Lee, Pil-Sung Jang, **Tae-Hwan Jin**
   Samsung Electronics Co., Ltd.

5. **KR 10-1612739(2016.04.08)**
   Distance measurement apparatus and method utilizing time to digital converter having scalable resolution
   Tae Wook Kim, **Tae-Hwan Jin**